Grounding quality measurement for orchestrated AI reasoning chains: evidence-binding rate as an evaluation primitive for orchestrator-in-the-loop agentic systems

Gil Raitses
aimez.ai

Abstract

Orchestrator-in-the-loop agentic systems route a user's request through a lead orchestrator that dispatches tool-bearing sub-agents and narrates a result, yet they ship without a metric for whether the narrated output is bound to the evidence the orchestrator actually gathered. R_{uncited} — the fraction of scientific sentences in a system's output that carry no bound citation — is a formal evaluation primitive for that orchestrated-reasoning quality. Using a reproducible 8-scenario parallel benchmark against the Gemini Interactions API with Google Maps grounding on marine science queries, Maps-only geospatial grounding achieves 60–100% uncited across query types. Injecting a structured orchestrated skill-dispatch step-log as RAG context reduces R_{uncited} to 0% by converting open-domain science queries into closed artifact-reference queries. The RAG lift hierarchy (step-log > structured data > narrow context > unstructured) diagnoses grounding architecture quality by context-type alignment. The metric applies directly to the orchestrator-in-the-loop architecture that contemporary human-agent systems adopt.

1 The evidence-binding gap in AI reasoning systems

Orchestrator-in-the-loop agentic systems route a user's request through a lead orchestrator that dispatches tool-bearing sub-agents and narrates a result. Magentic-UI is a representative instance: a lead orchestrator directs sub-agents and Model Context Protocol tools with the human in the loop through co-planning, action guards, and memory [20]. This realizes, at agent scale, the mixed-initiative principle Horvitz set out, coupling automated services with the user under explicit uncertainty about intent [14]. Treating the human and the orchestrator as communicative agents distinguished only by function, agency, interactivity, and influence, is the agent-agnostic stance human-machine communication adopts for exactly this configuration [3, 34], with agency and control continually negotiated between human and system rather than fixed [11]. Magentic-UI is evaluated on task completion, interaction cost, and safety; it proposes no metric for whether the orchestrator's narrated output is bound to the evidence its sub-agents actually gathered. Because users treat such systems as social actors [8] and accept fluent narration through heuristic rather than systematic processing [16], an unbound claim can pass human review unchallenged, and calibrated reliance instead depends on signals the narration does not expose [13]. That missing measurement is the evidence-binding gap, and it is the gap this paper defines and measures.

Scientific and physical-domain outputs make claims requiring empirical grounding. On orca encounter likelihood at Haro Strait, a language model may name the salmon corridor, acoustic monitoring network, and behavioral studies, without binding any claim to generating observations, prior analyses, or data artifacts. The response is fluent; it is not evidentially accountable. Fluency is not comprehension, and whether such systems understand what they emit is itself contested [19]. Fluent interfaces routinely manufacture an impression of accountability they have not earned, a banal deception embedded in everyday human-machine contact [21], and critical scholarship holds that such accountability gaps demand scrutiny beyond task-completion metrics [10]. An output-level binding criterion sidesteps the comprehension dispute by measuring traceability rather than adjudicating internal understanding.

Formally, the gap is the structural disconnection between stated claims and the observations, methods, and experiments that would make them verifiable, not factual accuracy per se, but whether a judge or downstream system can trace the claim to ground-truth data versus parametric pattern reproduction.

Prior work on RAG hallucination reduction [7, 9, 28, 24, 22] establishes that retrieval-augmented generation reduces factual error rates and that citation generation can be automated. However, this literature focuses on whether RAG improves accuracy, not on whether the improvement comes from evidence binding, from the system's outputs being traceable to their generating context, versus improved parametric knowledge activation.

The geospatial grounding baseline makes this distinction concrete. Google Maps grounding achieves high citation density for place-of-interest queries. For marine science evidence queries, the same domain, but requiring scientific rather than place-level grounding, Maps grounding resolves 0 of 25 citations to scientific or dataset sources. NOAA fisheries data, the Center for Whale Research census, Chinook salmon migration corridors, and acoustic hydrophone recordings are all generated as claims but remain entirely uncited. The system knows about these phenomena from training; it cannot bind its claims to verifiable data artifacts.

R_{uncited}, the fraction of scientific sentences in a system's output with no bound citation, is the measurement primitive that makes this gap quantitative, reproducible, and architecture-diagnostic.

2 R_{uncited} as a formal evaluation metric

2.1 Definition

Let S be the set of sentences in a system's output. Define $S_{\text{sci}} \subseteq S$ as the sentences containing scientific or quantitative markers. For each sentence $s \in S_{\text{sci}}$, let $\text{cited}(s)$ be the indicator that s names at least one entity appearing in the bound citation set C.

$$R_{\text{uncited}} = \frac{|\{s \in S_{\text{sci}} : \neg\, \text{cited}(s)\}|}{|S_{\text{sci}}|} \tag{1}$$

$R_{\text{uncited}} = 0$ when every scientific sentence names a bound citation. $R_{\text{uncited}} = 1$ when no scientific sentence does. It is undefined when $|S_{\text{sci}}| = 0$.

2.2 Properties

R_{uncited} has four properties that make it useful as an evaluation primitive.

Claim-level granularity. The metric operates at the sentence level. A response with 10 scientific sentences and 9 bound citations still has $R_{\text{uncited}} = 0.1$, distinguishing it from response-level pass/fail metrics.

Architecture-diagnostic. Two systems with the same surface accuracy can have very different R_{uncited} values depending on whether their reasoning traces are passed as context to the language model. The metric discriminates between evidence-bound and parametric-knowledge-based claim generation.

Reproducible. R_{uncited} is computed from the raw API output. No human annotation is required. The scientific marker set is implemented as a regex over domain-specific technical terms.

Model-independent. R_{uncited} can be applied to any language model's output. It measures the binding between output and context, not internal model behavior.

2.3 Relation to prior evaluation frameworks

The Ragas framework [7] evaluates faithfulness (whether claims are supported by retrieved context), answer relevancy, and context precision. These metrics require a reference context. R_{uncited} measures a weaker but more deployable condition: whether the model's output names sources in the context it was given, without requiring a predefined reference set.

The ALCE benchmark [9] evaluates citation generation at the claim level. R_{uncited} is complementary: it measures the rate of claims carrying no citation at all, regardless of citation validity.

R_{uncited} is also distinct from explainable-AI evaluation, which concerns the transparency of a model's decision process rather than the binding of its output to gathered evidence [5], and from reasoning-chain evaluation, which scores the internal correctness and informativeness of intermediate steps rather than their traceability to artifacts [25].

3 Benchmark methodology and results

3.1 Setup

The benchmark uses the Gemini Interactions API (gemini-3.5-flash, Api-Revision: 2026-05-20) with `google_maps` tool at San Juan Islands coordinates (48.5465, −123.03). Eight scenarios run in parallel via Python asyncio (`tools/testing/grounding_parallel_rag.py`), each submitting a distinct query with a distinct RAG context type. Run date: 2026-06-24.

3.2 Scenarios

#	Context injected	Query type
1	Maps-only (none)	Where are SRKWs seen + evidence
2	Live gate data	Same + gate framing
3	Cell provenance	Cell intensity + kernels
4	Surface planner step-log	Gates blocking promotion
5	Maps-only (none)	Sighting check
6	Sighting-assist output	Same + orcast data
7	Maps-only (none)	Shore whale watching plan
8	Hotspot probability data	Same + orcast forecast

3.3 Results

Scenario	Context	R_{uncited}
4	Planner step-log	**0%**
8	Hotspot data	75%
1	Maps-only orca	60%
7	Maps-only trip	100%
2	Gate context	100%
3, 5, 6	Provenance / sighting	100%

Scenario 4 achieves $R_{\text{uncited}} = 0\%$. Scenario 8's hotspot data lowers R_{uncited} by 25 pp relative to the same trip-planning query without it (Scenario 7, 100%), partially substituting for science claims.

4 The RAG lift hierarchy as a grounding architecture diagnostic

The 8-scenario results reveal a diagnostic structure:

Context type	R_{uncited}
Step-log (complete reasoning trace)	0%
Structured domain data (hotspots)	75%
Maps-only baseline	60–100%
Narrow structured data (gates)	100%

This is not a function of data volume but of structural alignment between context type and the science claims the model otherwise generates.

Full grounding (step-log): The planner step-log records skills dispatched, panels selected, and artifacts referenced; the query is answerable from those references without open-domain science claims. R_{uncited} collapses to 0% because $|S_{\text{sci}}| = 0$.

Partial grounding (hotspot data): Structured probability data substitutes for some generic science claims; the remaining $R_{\text{uncited}} = 75\%$ is ecology claims the hotspot data does not cover.

No grounding (gate data): Gate data covers statistical test outcomes only; the model still generates broader ecology claims gate data does not address.

Context type is a grounding affordance that shapes which claims become evidence-bound [6]; complete reasoning traces eliminate uncited claims; narrow structured context does not. The diagnostic ports to any domain where the reasoning trace can be injected as context; eRAG retrieval evaluation [30] would show the same pattern, only step-log documents are fully utilized.

The Citation-Enhanced Generation framework [27] demonstrates that post-hoc citation binding reduces hallucination in chatbots. The hierarchy finding complements this: pre-hoc trace injection eliminates the class of queries that citation binding must address.

5 Extension to orchestrated and planning systems

5.1 The step-log as a planning trace

Scenario 4's surface planner step-log is a planning trace that records the query, the skills dispatched, and the intermediate states. Injected as RAG context, the model answers from trace artifact references rather than world knowledge; $R_{\text{uncited}} = 0\%$ because the query becomes closed artifact-reference, not open-domain science.

The paper "Reasoning with Language Model is Planning with World Model" [12] explicitly frames LLM reasoning steps as world-state traces and proposes evaluating whether those states are accurate. The evaluation problem R_{uncited} addresses is the evidence-binding dimension of this: not whether the intermediate states are accurate, but whether each state is traceable to the observations that produced it. TRACE [33] demonstrates that constructing knowledge-grounded reasoning chains from structured context eliminates irrelevant retrieval noise in multi-hop queries; the orcast step-log is this structure applied to skill-dispatch interactions. Chain-of-Thought evaluation [35] confirms that intermediate reasoning steps are causally important for correct final outputs, supporting the claim that step-log quality measurement is foundational to planning system evaluation.

5.2 Applying R_{uncited} to orchestrated agentic systems

Section 4's benchmark applies to any orchestrator-in-the-loop agentic system that produces a structured interaction trace. Magentic-UI exemplifies this class [20]: inject the orchestration trace as RAG context, ask a domain-specific question, and measure R_{uncited}, bound artifact references drive R_{uncited} toward 0%; sparse traces leave it high. As users are increasingly asked to accept and rely on such systems [17], the metric supplies the output-grounding measurement that orchestrator-in-the-loop systems are currently deployed without.

6 Limits and falsification

6.1 Scope limits

Domain specificity. The benchmark uses marine science queries. Extension to other domains requires updating the science marker regex for that domain's technical vocabulary.

Model dependence. Results were produced with gemini-3.5-flash (2026-06-24). A model with stronger domain-specific knowledge may generate fewer science claims.

R_{uncited} does not measure semantic accuracy. A claim that is cited to a bound artifact but semantically wrong is not captured. Extension to factual accuracy requires a reference ground truth.

Step-log quality dependence. $R_{\text{uncited}} = 0\%$ requires a complete, structured step-log; sparse or malformed traces would not achieve the same result.

6.2 Falsification conditions

Claim	Falsifying condition
Step-log injection achieves $R_{\text{uncited}} = 0\%$	Step-log-injected query produces uncovered science claims
Hierarchy is stable across query types	Different Scenario 4 query yields high R_{uncited}
Gate context achieves no lift	Gate data injection covers generated science claims
R_{uncited} diagnoses architecture quality	Two architecturally different systems produce identical hierarchies

6.3 Extension path

Multi-domain validation: run the benchmark on pedestrian routing (pax), clinical care, and manufacturing. World model evaluation: apply the benchmark to JEPA-based world models (V-JEPA [4]) and measure whether planning trace injection reduces R_{uncited}. Semantic accuracy addition: add a second metric for whether cited claims are factually accurate relative to their bound artifacts.

6.4 The benchmark gap in existing physical-world evaluation

Physical-world AI benchmarks evaluate task completion or prediction accuracy without auditing intermediate claim binding. PHYRE [2] measures physical reasoning via puzzle solve rate. STAR [32] evaluates situated reasoning in real-world videos via question answering accuracy. ThreeDWorld [23] measures embodied planning by object transport success rate. TravelPlanner [36] evaluates language agent planning by plan validity. The comprehensive Embodied AI survey [26] categorizes evaluation as navigation, interaction, manipulation, and reasoning accuracy. None measures whether intermediate reasoning claims are bound to their generating sensor observations. EWOK [29] tests world modeling via plausibility matching. The need for evaluation primitives beyond functional task success has a precedent: human-robot interaction required psychological benchmarks to capture what task-completion metrics omit [15]. R_{uncited} provides the missing primitive for evidence binding.

6.5 World model evaluation as a future extension

A natural extension is to apply this benchmark to world model planning systems whose design explicitly requires sensor-grounded intermediate representations. LeCun's autonomous machine intelligence architecture [18] specifies that the world model module must learn to represent the information content of the percept that is relevant for predicting future states, a grounding requirement on intermediate representations. Joint-embedding predictive architectures, from I-JEPA [1] to V-JEPA [4], and Image World Models [31] are evaluated on downstream accuracy on image and video benchmarks; no claim-level evidence binding metric is proposed. Whether R_{uncited} on JEPA planning traces reproduces Section 5's diagnostic hierarchy is an open empirical question. Section 4's methodology requires only a structured planning trace; running it on world model outputs needs no architectural change.

References

[1] M. Assran et al. *Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture*. 2023. arXiv: 2301.08243. URL: https://arxiv.org/abs/2301.08243.

[2] A. Bakhtin et al. *PHYRE: A New Benchmark for Physical Reasoning*. 2019. arXiv: 1908.05656. URL: https://arxiv.org/abs/1908.05656.

[3] J. Banks and M. M. A. de Graaf. "Toward an Agent-Agnostic Transmission Model: Synthesizing Anthropocentric and Technocentric Paradigms in Communication". In: *Human-Machine Communication* 1 (2020), pp. 19–36. DOI: 10.30658/hmc.1.2.

[4] A. Bardes et al. *Revisiting Feature Prediction for Learning Visual Representations from Video*. 2024. arXiv: 2404.08471. URL: https://arxiv.org/abs/2404.08471.

[5] M. A. Clinciu and H. Hastie. "A Survey of Explainable AI Terminology". In: *Proceedings of the 1st Workshop on Interactive Natural Language Technology for Explainable Artificial Intelligence*. Association for Computational Linguistics, 2019, pp. 8–13.

[6] J. L. Davis. "Affordances for Machine Learning". In: *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*. ACM, 2023, pp. 324–332. DOI: 10.1145/3593013.3594000.

[7] S. Es et al. *Ragas: Automated Evaluation of Retrieval Augmented Generation*. 2023. arXiv: 2309.15217. URL: https://arxiv.org/abs/2309.15217.

[8] A. J. Gambino, J. Fox, and R. A. Ratan. "Building a Stronger CASA: Extending the Computers Are Social Actors Paradigm". In: *Human-Machine Communication* 1 (2020), pp. 71–85. DOI: 10.30658/hmc.1.5.

[9] T. Gao et al. *Enabling Large Language Models to Generate Text with Citations*. 2023. arXiv: 2305.14627. URL: https://arxiv.org/abs/2305.14627.

[10] P. M. Gardner and J. Rauchberg. "Feminist Cybernetic, Critical Race, Postcolonial, and Crip Propositions for the Theoretical Future of Human-Machine Communication". In: *Human-Machine Communication* 8 (2024), pp. 27–51. DOI: 10.30658/hmc.8.2.

[11] J. L. Gibbs, G. L. Kirkwood, C. Fang, and J. N. Wilkenfeld. "Negotiating Agency and Control: Theorizing Human-Machine Communication From a Structurational Perspective". In: *Human-Machine Communication* 2 (2021), pp. 153–172. DOI: 10.30658/hmc.2.8.

[12] S. Hao et al. *Reasoning with Language Model is Planning with World Model*. 2023. arXiv: 2305.14992. URL: https://arxiv.org/abs/2305.14992.

[13] K. A. Hoff and M. Bashir. "Trust in Automation: Integrating Empirical Evidence on Factors That Influence Trust". In: *Human Factors* 57.3 (2015), pp. 407–434. DOI: 10.1177/0018720814547570.

[14] E. Horvitz. "Principles of Mixed-Initiative User Interfaces". In: *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '99)*. 1999, pp. 159–166. DOI: 10.1145/302979.303030.

[15] P. H. Kahn et al. "What is a Human? Toward psychological benchmarks in the field of human-robot interaction". In: *Interaction Studies* 8.3 (2007), pp. 363–390. DOI: 10.1075/is.8.3.04kah.

[16] K. Koban and J. Banks. "Dual-Process Theory in Human-Machine Communication". In: *Communication*. SAGE Publications Ltd, 2023. DOI: 10.4135/9781529782783.

[17] P. D. Koenig. "Attitudes toward Artificial Intelligence: Combining Three Theoretical Perspectives on Technology Acceptance". In: *AI & Society* 40 (2025), pp. 1333–1345. DOI: 10.1007/s00146-024-01987-z.

[18] Y. LeCun and K. Alahari. *Introduction to Latent Variable Energy-Based Models: A Path Towards Autonomous Machine Intelligence*. 2023. arXiv: 2306.02572. URL: https://arxiv.org/abs/2306.02572.

[19] M. Mitchell and D. C. Krakauer. "The debate over understanding in AI's large language models". In: *Proceedings of the National Academy of Sciences* 120.13 (2023), e2215907120. DOI: 10.1073/pnas.2215907120.

[20] H. Mozannar, G. Bansal, C. Tan, A. Fourney, et al. *Magentic-UI: Towards Human-in-the-loop Agentic Systems*. Microsoft Research AI Frontiers, MSR-TR-2025-40. 2025. arXiv: 2507.22358. URL: https://arxiv.org/abs/2507.22358.

[21] S. Natale. *Deceitful Media: Artificial Intelligence and Social Life after the Turing Test*. Oxford University Press, 2021.

[22] S. J. Semnani et al. *WikiChat: Stopping the Hallucination of Large Language Model Chatbots by Few-Shot Grounding on Wikipedia*. 2023. arXiv: 2305.14292. URL: https://arxiv.org/abs/2305.14292.

[23] Unknown. *The ThreeDWorld Transport Challenge: A Visually Guided Task-and-Motion Planning Benchmark for Physically Realistic Embodied AI*. 2021. arXiv: 2103.14025. URL: https://arxiv.org/abs/2103.14025.

[24] Unknown. *AGREE: Achieving Ground Truth Faithful Generations with LLMs*. 2023. arXiv: 2311.09533. URL: https://arxiv.org/abs/2311.09533.

[25] Unknown. *ReCEval: Evaluating Reasoning Chains via Correctness and Informativeness*. 2023. arXiv: 2304.10703. URL: https://arxiv.org/abs/2304.10703.

[26] Unknown. *Aligning Cyber Space with Physical World: A Comprehensive Survey on Embodied AI*. 2024. arXiv: 2407.06886. URL: https://arxiv.org/abs/2407.06886.

[27] Unknown. *Citation-Enhanced Generation for LLM-based Chatbots*. 2024. arXiv: 2402.16063. URL: https://arxiv.org/abs/2402.16063.

[28] Unknown. *CRAG – Corrective Retrieval Augmented Generation*. 2024. arXiv: 2401.15884. URL: https://arxiv.org/abs/2401.15884.

[29] Unknown. *Elements of World Knowledge (EWOK): A Cognition-Inspired Framework for Evaluating Basic World Knowledge in Language Models*. 2024. arXiv: 2405.09605. URL: https://arxiv.org/abs/2405.09605.

[30] Unknown. *Evaluating Retrieval Quality in Retrieval-Augmented Generation*. 2024. arXiv: 2404.13781. URL: https://arxiv.org/abs/2404.13781.

[31] Unknown. *Learning and Leveraging World Models in Visual Representation Learning*. 2024. arXiv: 2403.00504. URL: https://arxiv.org/abs/2403.00504.

[32] Unknown. *STAR: A Benchmark for Situated Reasoning in Real-World Videos*. 2024. arXiv: 2405.09711. URL: https://arxiv.org/abs/2405.09711.

[33] Unknown. *TRACE the Evidence: Constructing Knowledge-Grounded Reasoning Chains for Retrieval-Augmented Generation*. 2024. arXiv: 2406.11460. URL: https://arxiv.org/abs/2406.11460.

[34] J. B. Walther. "Theories of Computer-Mediated Communication and Interpersonal Relations". In: *The Handbook of Interpersonal Communication*. Ed. by M. L. Knapp and J. A. Daly. 4th ed. SAGE, 2011, pp. 443–479.

[35] J. Wei et al. *Chain of Thought Prompting Elicits Reasoning in Large Language Models*. 2022. arXiv: 2201.11903. URL: https://arxiv.org/abs/2201.11903.

[36] J. Xie et al. *TravelPlanner: A Benchmark for Real-World Planning with Language Agents*. 2024. arXiv: 2402.01622. URL: https://arxiv.org/abs/2402.01622.